

SOMIDEM TABLET 10 MG

DESCRIPTION:

SOMIDEM TABLET 10 MG: An 11 mm white caplet with 'DUO' marking.

COMPOSITION:

Each tablet contains Zolpidem Tartrate 10 mg

PHARMACODYNAMICS:

Zolpidem is a potent agonist with high intrinsic activity at the omega (ω) 1 subtype (also called the benzodiazepine 1 [BZ1] subtype) of the gamma-aminobutyric acid type (GABAA) receptor-chloride ionophore complex. The receptor complex resides on the neuronal membranes and functions in the gating of the chloride channel, allowing the flow of chloride ions through the neuronal membrane and into the neuron. This results in hyperpolarization, which inhibits the firing of that neuron.

In contrast to benzodiazepines, which bind non-selectively to the omega 1, omega 2 and omega 3 GABAA receptors, zolpidem possesses relative selectivity for the omega 1 GABAA receptor. This preference for the omega 1 GABAA receptor may account for zolpidem's relative lack of anticonvulsant, myorelaxant, and anxiolytic effects at therapeutic doses, and for the general preservation of sleep architecture seen with zolpidem use.

PHARMACOKINETICS:

Absorption: Rapid and complete, although first-pass metabolism results in 70% bioavailability. Food may decrease the rate and extent of absorption.

Distribution: The volume of distribution (V₀D) of zolpidem in healthy volunteers was 0.54 L per kg (L/kg) following an 8 mg intravenous dose. Zolpidem is distributed into breast milk; amounts ranging from 0.004 to 0.019% of a 20-mg oral dose were present in milk samples taken 3 hours following administration.

Protein binding: Very high (92%).

Biotransformation: Hepatic, resulting in 3 major and several minor metabolites.

Half-life: Elimination - 2.6 hours (range, 1.4 to 4.5 hours). The elimination half-life of zolpidem is prolonged in the elderly and in patients with impaired hepatic or renal function.

Onset of action: Rapid.

Time to peak concentration: 30 minutes to 2 hours; may be longer if zolpidem is taken with food.

Peak serum concentration: Mean peak plasma concentration (C_{max}) following oral administration of 5 mg of zolpidem in healthy volunteers was 59 nanograms per mL (nanograms/mL) (0.077 micromoles per L [micromoles/L]), with a range of 29 to 113 nanograms/mL (0.038 to 0.148 micromoles/L); C_{max} following administration of 10 mg of zolpidem was 121 nanograms/mL (0.158 micromoles/L) with a range of 58 to 272 nanograms/mL (0.076 to 0.356 micromoles/L).

Elimination:

Renal - 48 to 67% of a single dose is eliminated in the urine. Fecal - 29 to 42% of a single dose is eliminated in the feces. Unchanged zolpidem is present in trace amounts in urine and feces. In dialysis - Not hemodialyzable.

INDICATION:

Zolpidem is indicated for short-term treatment of insomnia. Hypnotics should generally be limited to 7 to 10 days of use, and re-evaluation of the patient is recommended if insomnia fails to remit after this period.

RECOMMENDED DOSAGE:

Posology

Dose

The treatment should be taken as a single dose and not be re-administered during the same night.

The recommended daily dose for adults is 10 mg to be taken immediately at bedtime.

Treatment should be started at the lowest effective dose and the maximum dose must not exceed 10 mg.

Elderly or debilitated subjects

As elderly or debilitated patients are especially sensitive to the effects of zolpidem, the recommended dose in this population is 5 mg (i.e. 1/2 a tablet).

Liver failure

As the clearance and metabolism of zolpidem are reduced in liver failure, treatment should begin at 5 mg per day in these patients, with particular caution being exercised in elderly patients.

Under no circumstances should the daily dose exceed 10 mg.

Somidem may be prescribed for use continuously or as required, depending on the signs and symptoms.

Pediatric population

The safety and efficacy of zolpidem in children and adolescents under 18 years of age have not been demonstrated. Therefore, use of zolpidem is not recommended in this population.

Duration

As with all hypnotics, long-term use of zolpidem is not recommended. Treatment should be as short as possible, and should not exceed four weeks, including the tapering-off period (see section Special warnings and special precautions for use).

The duration of treatment must be explained to the patient:

- 2 to 5 days for transient insomnia (when traveling, for example),
- 2 to 3 weeks for short-term insomnia (following a serious event, for example).

Very short-term treatments do not require a tapering-off period.

Extension beyond the maximum treatment period should not take place without re-evaluation of the patient's status, since the risk of abuse and dependence increases with the duration of treatment.

ROUTE OF ADMINISTRATION:

For oral use

CONTRAINDICATIONS:

- Sensitivity to Zolpidem.
- Alcohol intoxication, acute, with depressed vital signs.
- Sleep apnoea, established or suspected.
- Severe respiratory insufficiency.
- Severe hepatic insufficiency.
- The use of this drug is generally inadvisable in the following situation: Children under age 18 years old, breast feeding, myasthenia.

WARNINGS & PRECAUTIONS:

THIS DRUG MAY BE HABIT FORMING ON PROLONGED USED

- Anaphylaxis (Severe allergic reaction) and angioedema (severe facial swelling), which can occur as early as the first time the product is taken.
- Complex Sleep – Related behaviors which may include sleep driving, making phone calls, preparing and eating food (while asleep).

Pediatrics: Studies on the relationship of age to the effects of zolpidem have not been performed in children up to 18 years of age. Safety and efficacy have not been established.

Geriatrics: It has been shown zolpidem to have an increased half-life, peak plasma concentration, and area under the plasma concentration time curve in geriatric patients. Elderly patients may be more likely to experience confusion or falls while taking zolpidem. A reduced starting dosage and careful monitoring are recommended. In addition, geriatric patients are more likely to have age-related renal function impairment, which may require dosage reductions. Patient should be cautioned against engaging in hazardous occupations requiring complete mental alertness or motor coordination such as operating machinery or driving a motor vehicle after ingesting zolpidem, including potential impairment of the performance of such activities that may occur the day following ingestion of zolpidem. Zolpidem showed additive effects when combined with alcohol and should not be taken with alcohol.

INTERACTION WITH OTHER MEDICAMENTS:

Alcohol or CNS depression-producing medications - Concurrent use may increase the CNS effects of either these medications or zolpidem caution is recommended, and dosage of one or both agents should be reduced.

Chlorpromazine - concurrent use may prolong elimination half-life of chlorpromazine.

Imipramine - concurrent use may increase drowsiness and incidence of anterograde amnesia, and decreased peak concentrations of imipramine.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Zolpidem has not been studied in pregnant women. Teratology studies were conducted in rats and rabbits. In rats, maternal lethargy, ataxia and under ossification of fetal bones occurred at 20 and 100 mg base/kg dose. In rabbits, dose-related maternal sedation, decreased weight gain occurred at all doses tested. Increase in post implantation fetal loss and under ossification of stern brae in viable fetuses occurred at dose of 16 mg base/kg. No frank teratogenic effects observed. Because animal reproduction studies are not always predictive of human response, zolpidem should be used during pregnancy only if clearly needed.

Use in Lactation: Nursing mothers showed < 0.02% of a single oral dose of zolpidem was distributed into breast milk. The effect of zolpidem on the infant is not known. In addition, in a rat study, zolpidem inhibited the secretion of milk. The use of zolpidem in nursing mothers is not recommended.

SIDE EFFECTS:

Ataxia, confusion - in the elderly: mental depression.

Allergic reaction or rash:

Anaphylaxis - fast heartbeat, swelling of face, wheezing or difficulty in breathing. **Falling** - in the elderly **Hypotension** - dizziness, light-headedness or fainting.

Paradoxical reactions, including agitation - unusual excitement or nervousness or irritability **Hallucinations** - seeing, hearing or feeling things that are not there, or insomnia - trouble in sleeping.

Gastrointestinal: Abdominal cramps and discomfort, gastric pain, diarrhoea, nausea, vomiting.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms:

Ataxia, severe (clumsiness or unsteadiness); cardiovascular compromise (slow heartbeat); diplopia (double vision) or disturbed vision; severe dizziness; severe drowsiness; severe nausea; respiratory problems (troubled breathing); unconsciousness; severe vomiting.

Treatment:

Treatment is essentially symptomatic and supportive, possibly including:

- Monitoring respiratory, cardiac, and CNS status and providing general supportive therapy as indicated.
- Inducing emesis or performing gastric lavage as appropriate.
- Administering activated charcoal to increase clearance and decrease absorption of zolpidem.
- Withholding sedating drugs even if excitation occurs.

Antidote:

Flumazenil may be useful in reversing zolpidem's sedative and respiratory depression effects.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE:

Blister pack of 10 x 10's tablets/ box

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

